N. 1

Dimostrare che $\forall n \geq 2$

$$F_n^2 - F_{n-1}F_{n+1} = (-1)^{n+1}$$

P. B.

$$F_2^2 - F_1 \cdot F_3 = (-1)^{n+1}$$

$$(1)^2 - (1)\cdot(2) = -1 = (-1)^{2+1} \quad \text{OK}$$

P. I.

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^{n+2}$$

$$(F_n + F_{n-1})^2 - F_n(F_{n+1} + F_n) = (-1)^{n+2}$$

$$\cancel{F_n^2} + 2F_nF_{n-1} + F_{n-1}^2 - F_nF_{n+1} - \cancel{F_n^2} = (-1)^n(-1)^2$$

$$2F_nF_{n-1} + F_{n-1}^2 - F_n(F_n + F_{n-1}) = (-1)^n(-1)^2$$

$$F_{n-1}(2F_n + F_{n-1} - F_n) - F_n^2$$

$$F_{n-1}(F_n + F_{n-1}) - F_n^2$$

$$F_{n-1}(F_{n+1}) - F_n^2 = (-1)^{n+2}$$

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^{n+2}$$

N. 2

$$[((p \land q) \to \neg r) \land (p \to (q \lor \neg r))] \to (p \to \neg r)$$

$((p \land q) \to \neg r) \land (p \to (q \lor \neg r))$

$\neg(p \to \neg r)$

$(p \land q) \to \neg r$

$p \to (q \lor \neg r)$

p

r

$\neg(p \land q)$	$\neg r$
$\neg p$	X
$\neg q$	
X	

Tautologia

- $[((p \land \neg q) \to r) \land (p \to (\neg q \lor r))] \to (p \to r)$

$\neg F$

$[((p \land \neg q) \to r) \land (p \to (\neg q \lor r))]$

$\neg(p \to r)$

$(p \land \neg q) \to r$

$p \to (\neg q \lor r)$

p

$\neg r$

$\neg(p \land \neg q)$	r
$\neg p$	X
q	
X	

Tautologia

N. 3

$$\forall_x [P(x) \vee Q(x)] \rightarrow \exists_x [P(x) \wedge Q(x)]$$

① $x : \mathbb{N}$

$P(x) : x \cdot y = 0 \qquad y \in \mathbb{N} \setminus \{0\}$

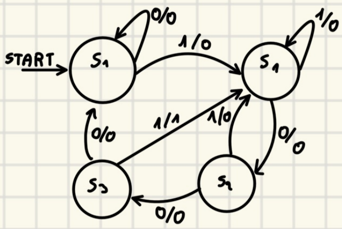
$Q(x) : x \cdot y = k \qquad y \in \mathbb{N} \setminus \{0\}$

② $\neg[\neg\exists_x \neg[P(x) \vee Q(x)]] \vee \exists_x [P(x) \wedge Q(x)]$

③ $\neg\{\neg[\forall_x[P(x) \vee Q(x)] \vee \neg\forall_x [P(x) \wedge Q(x)]]\}$

N. 4

t	1	2	3	4	5	6	7	8
x	0	0	1	0	0	1	0	0

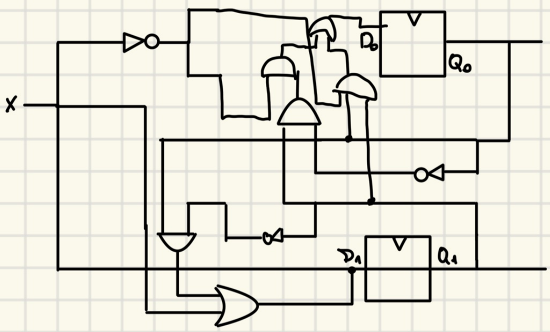


Q_0	Q_1	x	D_0	D_1	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

x \ Q_0Q_1	00	01	11	10
0	0			1
1	1	1	1	1

$D_1 = x + Q_0 \overline{Q_1}$

$D_0 = \overline{Q_0} Q_1 \overline{x} + Q_0 \overline{Q_1}\, \overline{x}$



N. 5

C3CAB0000

$\underbrace{1}\,\underbrace{1\;0\;0\;0\;0\;1\;1\;1}\;1\;0\;0\;1\;0\;1\;0\;1\;0\;1\;1\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0\;0$

$2^0 + 2^1 + 2^2 + 2^7 = 1 + 2 + 4 + 128 = 135 - 127 = 8$

$1.000011110010101011 \cdot 10^8$

$=$

100001111.0010101011

$2^0 + 2^1 + 2^2 + 2^3 + 2^8 =$

$1 + 2 + 4 + 8 + 256 = 271,\ 2^{-2} + 2^{-4} + 2^{-6} + 2^{-8} + 2^{-9}$

$-271,\ \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \frac{1}{512}$